Recipe Knowledge Graph
& GraphRAG System

Social Media Analytics - Final Project

Recipe | Ingredient | Cuisine | User | Review

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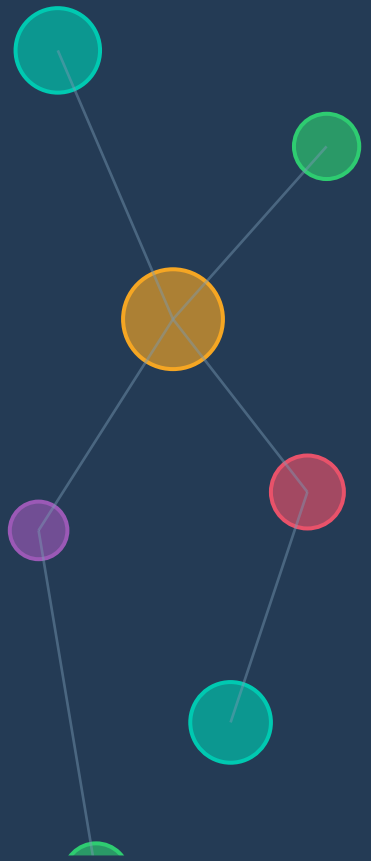


Data Preprocessing → LLM Knowledge Extraction → Neo4j Graph DB → GraphRAG Q&A

Team Members

Suryah

Preprocessing & EDA

Nandini

Knowledge Graph

Varsha

Neo4j Graph Database

Sarvesh

GraphRAG Developer

Riju

Streamlit

The 4-Step Pipeline

1 Data Preprocessing

- Load & clean CSV files
- Normalize ingredients
- Merge recipes + interactions

↳ master_dataset.parquet

2 Knowledge Graph (LLM)

- Design KG schema
- LLM triple extraction
- Ingredient substitutions

↳ kg_triples.json / kg_nodes.json

3 Neo4j Storage

- Constraints & indexes
- MERGE nodes & edges
- Verify with Cypher

↳ Populated graph DB

4 GraphRAG Q&A

- Parse user question
- Build Cypher query
- LLM generates answer

↳ answer_question() fn

5 Integration & Demo

Streamlit app · End-to-end pipeline test

PHASE 1

Data Preprocessing

recipes.csv

- id, name, minutes, contributor_id
- submitted, tags, nutrition
- n_steps, steps, description
- ingredients, n_ingredients

+

interactions.csv

- user_id, recipe_id
- date, rating (0–5)

→

master_dataset .parquet

Cleaning Tasks

T1.1
Load CSVs, parse list columns (tags, ingredients, nutrition)

T1.2
Remove outliers, normalize ingredients, extract cuisine & diet from tags

T1.3
Merge, compute avg_rating, produce master_dataset.parquet

231K	1.1M+	35+	21
Recipes	Reviews	Cuisines	Diet Flags

PHASE 2

Knowledge Graph Schema

8 Node Types

Recipe	id · name · minutes · avg_rating
Ingredient	name
Cuisine	name
DietaryRestriction	name
Tag	name
NutritionProfile	calories · protein · fat · carbs
User	id · role
Review	text · rating · date

10 Relationship Types

USES_INGREDIENT	Recipe → Ingredient	LLM
BELONGS_TO_CUISINE	Recipe → Cuisine	Rules
FITS_DIET	Recipe → DietaryRestriction	Rules
HAS_TAG	Recipe → Tag	LLM
HAS_NUTRITION	Recipe → NutritionProfile	Data
CONTRIBUTED_BY	Recipe → User	Data
HAS_REVIEW	Recipe → Review	Data
WRITTEN_BY	Review → User	Data
RATES	Review → Recipe	Data
SUBSTITUTES_FOR	Ingredient → Ingredient	LLM

PHASE 2

Knowledge Graph — Network View

2,282 nodes across 8 types: Recipe, Ingredient, Cuisine, DietaryRestriction, Tag, NutritionProfile, User, Review

5,647+ triples covering ingredients, cuisines, diet flags, tags, nutrition, reviews and substitutions

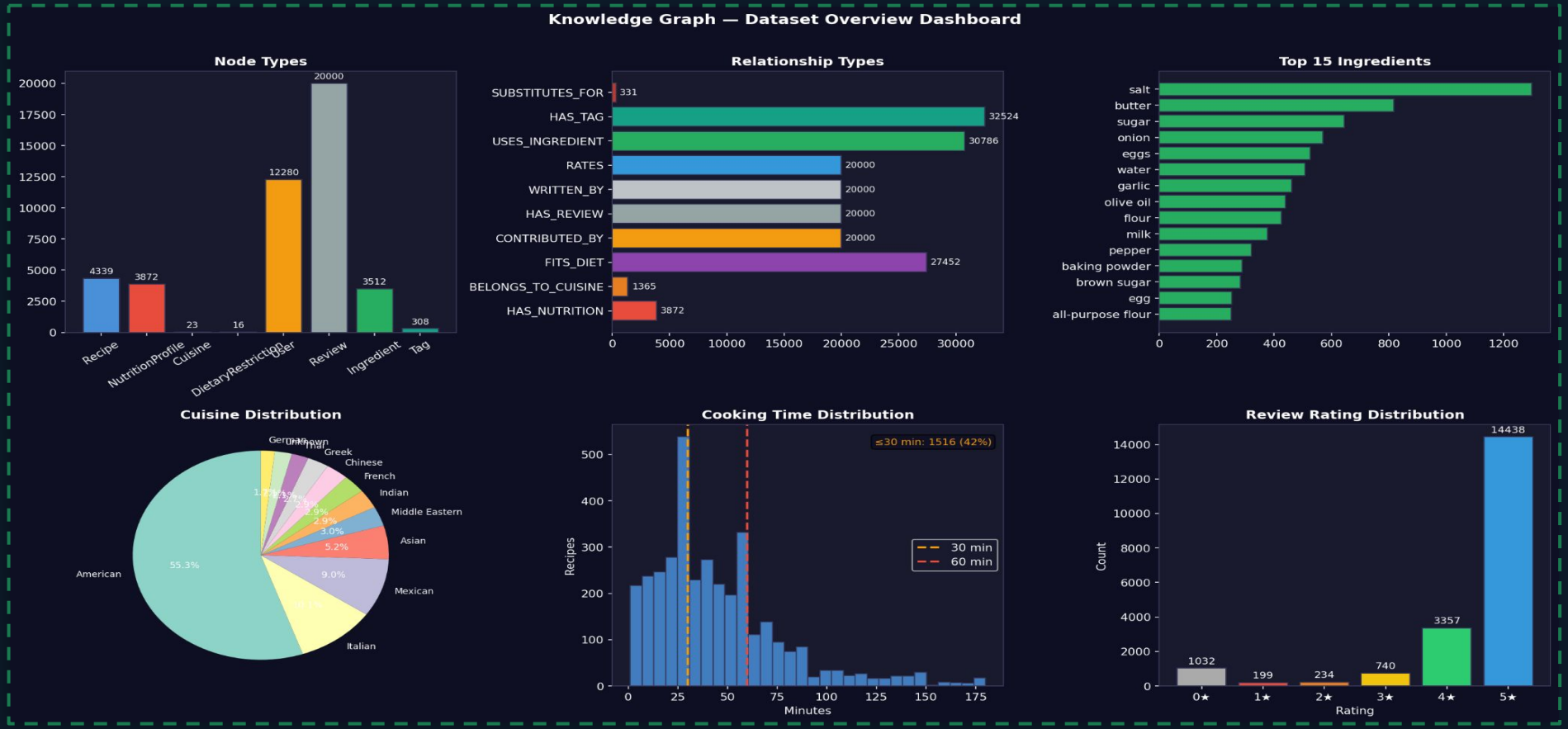
Every edge carries a minutes property — enabling time-based filtering directly in Cypher without touching node properties

300 SUBSTITUTES_FOR triples generated by the LLM across the top 100 most common ingredients — pure knowledge not in any dataset column

2,282+	5,647+	8	10	≤30min
Nodes	Triples	Node Types	Rel. Types	Time filter

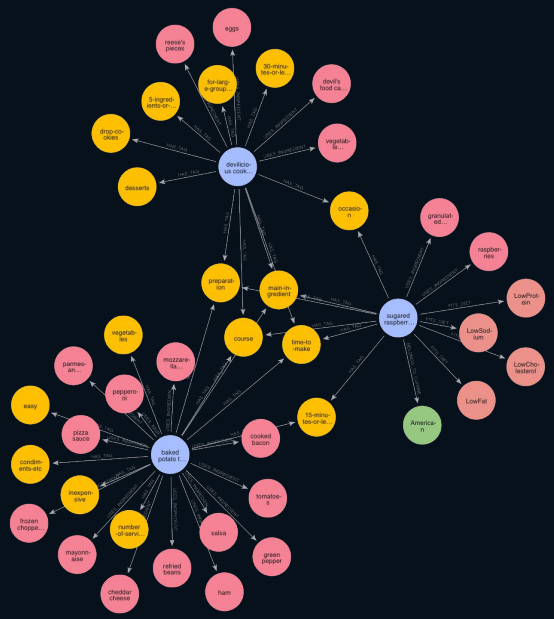
PHASE 2

Dataset Analytics Dashboard



PHASE 3

Neo4j Graph Database



Sample Cypher Queries

Generate the Sample Graph

```
MATCH
(r:Recipe)-[:USES_INGREDIENT]->(i:Ingredient)
WITH r, collect(i) AS ingredients
WHERE size(ingredients) >= 6
LIMIT 5
UNWIND ingredients AS ing
MATCH (r)-[rel]->(n)
RETURN r, rel, n
LIMIT 120;
```

Quick recipes ≤30m

```
MATCH (r:Recipe)
WHERE r.minutes <= 30
RETURN r.name
LIMIT 10
```

Role of Neo4j in this project

- Efficient traversal of complex relationships
- Natural representation of recipe-ingredient networks
- Enables semantic search and recommendations
- Supports GraphRAG integration with LLMs

PHASE 4

GraphRAG Q&A Pipeline

User Question → LLM Entity Extraction → Cypher Query Builder → Neo4j Lookup → LLM Answer Generation

"Mexican cuisine using onion and green chilies under 60 minutes"
"Vegetarian Italian dishes under 30 minutes"

① Extracted Entities

```
diet: Vegetarian
cuisine: Italian
max_minutes: 30
type: find_recipe
```

② Cypher Query

```
MATCH (r:Recipe)-[e]->
(c:Cuisine)
WHERE c.name='Italian'
AND e.minutes <= 30
```

③ Natural Language Answer

```
"Based on the graph,
5 Italian recipes
under 30 min..."
(grounded in real data)
```

PHASE 5

Live Demo — Streamlit Application

📷 demo_app.py — Streamlit running with a real query (user populated ingredients) and answer

Filters

Max prep time (minutes)

60

5 180

Min matched ingredients

2

1 10

Reset filters

Deploy

🍽️ **Recipe Discovery with GraphRAG**

Discover recipes with graph-based retrieval and grounded AI explanations. Enter ingredients or ask naturally.

Try one of these starter prompts

chicken, garlic, spinach

what can I make with potatoes and cheese

milk, butter, salt

quick breakfast with eggs

Search query

chicken, garlic, spinach

Search

Top matches for 'chicken, garlic, spinach': spinach and chile stir fry, fresh spinach salad, artichoke and pasta salad, spinach dish, north african chicken.

Recipes	Avg Time	Avg Matches
8	27 min	2.0

PHASE 5

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Reset filters

Deploy

Recipes	Avg Time	Avg Matches
8	27 min	2.0

1. Spinach And Chile Stir Fry

Cuisine: American **Time:** 15 min **Matched Ingredients:** 2

Key Ingredients: Fresh Cilantro Leaves | Salt | Garlic | Pepper | Fresh Lime Juice | Canola Oil | Spinach | Fresh Jalapeno Chilies | Vegan Sugar

How to make it:

1. Heat oil in large skillet
2. Add spinach and cilantro
3. Stir until spinach wilts , about 1 minute
4. Add jalapeno , sugar , garlic , and lime juice
5. Season to taste with salt and pepper
6. Cook until spinach is tender , another 2-3 minutes

Details

PHASE 5

Live Demo — Streamlit Application

📷 demo_app.py — Streamlit running with a natural language query and returning results

Deploy ⋮

Filters

Max prep time (minutes)

60

5 180

Min matched ingredients

2

1 10

Reset filters

Search query

what breakfast can I make with milk and eggs

Search

Top matches for 'what breakfast can I make with milk and eggs': jim s microwave scrambled eggs, sunny orange pancakes, my family pancakes, perfect corn dogs, down home breakfast special.

Recipes	Avg Time	Avg Matches
10	18 min	2.0

1. Jim S Microwave Scrambled Eggs

Cuisine: None **Time:** 9 min **Matched Ingredients:** 2

Key Ingredients: Eggs Onion Salt And Pepper Milk

How to make it:

1. Beat eggs well
2. Add milk , salt and pepper

APP LINK

Recipe GraphRAG

THANK YOU